Customer Stories / Gaming / Ukraine

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Quadrupling Bets per Second While Modernizing on AWS with Playson

Learn how casino game developer Playson enhanced the performance of its gaming service by going all in on AWS.

Overview | **Opportunity** | **Solution** | **Outcome** | **AWS Services Used**

200%

increase in bets per second

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Virtually zero downtime

during cloud migration

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Overview

For omnichannel game provider Playson, delivering games at low latencies is key to keeping players engaged and maximizing their betting opportunities. Since its founding, the company has quickly grown, expanding to new customer bases while regularly releasing fresh titles. To keep up with demand, the Playson team needed to scale its gaming service, which was difficult with its monolithic, multicloud architecture.

To enhance its agility, Playson decided to go all in with one cloud service provider and chose Amazon Web Services (AWS) for its scalability and reliability. In 8 months, the Playson team migrated its workloads from 2,000 servers to AWS. As a result, Playson can deliver a faster, more reliable gaming experience to players while simplifying its monitoring and governance capabilities.

In the next year, Playson fully rebuilt its gaming service, xPlatform, with a microservices architecture using Amazon Elastic Kubernetes Services (Amazon EKS), a managed Kubernetes service that runs Kubernetes on AWS and on-premises data centers. By taking a microservices approach, the company has reduced spin time and improved scalability during prime gaming times, which has led to an increase in bets of up to 200 percent per second.

Opportunity | Enrolling in AWS MAP to Modernize a Monolithic Architecture for Playson

Playson offers a portfolio of 69 cutting-edge, high-quality casino games, including titles like Royal Coins 2 and Lion Gems. The company provides its games to thousands of iGaming businesses, which can connect with Playson's turnkey gaming service through direct integrations. Playson has seen these collaborations grow by 8 percent year over year since 2015.

With its monolithic architecture, Playson would set up one server for each client it acquired. As the company grew, it became difficult to sustain this infrastructure model. Further, the company experienced downtime each time it released a title, which impacted the player experience. “Most players compare their online experience with playing slot machines in a casino,” says Oleksii Mylotskyi, chief technology officer at Playson. “So, everything that we offer has to be blazingly fast.”

In addition to its on-premises environment, Playson had a portion of its workloads running on services from multiple cloud providers, including AWS. As the company continued to grow, it wanted to achieve high availability and scalability by decoupling its applications from each other. “We wanted to transition to a microservices architecture and extensively use AWS services, with the intention of improving our system’s resilience, agility, and scalability,” says Mylotskyi. “Our desire was to break up our applications into smaller, autonomous services to facilitate a more efficient development cycle, provide superior fault isolation, and prompt feature releases.”

Playson understood the strategic advantage that fully committing to AWS could offer to its service. The comprehensive suite of AWS services, coupled with robust and scalable infrastructure, presented an opportunity to significantly enhance the value that Playson could provide to its customers. To plan for a zero-downtime migration, the company enrolled in the AWS Migration Acceleration Program (AWS MAP), which helps accelerate businesses’ migration and modernization journeys with an outcome-driven methodology.

Solution | Reducing Spin Time and Increasing Betting by 200% with a Microservices Architecture on AWS

The Playson team learned about AWS best practices through AWS MAP, helping it design an architecture that lent itself to low-latency requirements with input from subject matter experts. “The AWS account managers and solutions architects oversee certain domains and customer bases,” says Oleksandr Rudenko, platform tribe leader at Playson. “They know how the gaming industry works and have given us good advice.” To avoid downtime, the Playson team and AWS team worked together to migrate each of the company’s 2,000 servers to the cloud one by one.

During the migration, Playson adopted Amazon CloudFront, a content delivery network servic
performance, security, and developer convenience, to deliver its games at scale, and now proc PI
requests using Amazon EKS. For storing its key-value data, the company uses Amazon Memor

(Amazon MemoryDB), a Redis-compatible, durable, in-memory database service for ultrafast performance. As an object database, Playson has adopted Amazon DynamoDB, a fast, flexible NoSQL database service for single-digit millisecond performance at any scale.

By building a microservices architecture on AWS, Playson has reduced its games' spin time by up to 350 milliseconds, which is in the 95th percentile. This reduction has meant a smoother, more immersive player experience. "We offer the opportunity for players to play our games faster now," says Rudenko. "We even added a turbospin feature because we have the technical capability."

Because players can partake in more rounds over a shorter period, they tend to bet more often, too. In some cases, Playson's clients saw the number of bets per second increase up to four times, increasing their revenue streams and overall profitability. Another benefit of its microservices architecture is that Playson no longer experiences downtime when launching new games, speeding up its time to market. "We can deliver games globally up to 10 times faster than before," says Mylotskyi.

Outcome | Enhancing Resiliency and Availability Using Multiple AWS Regions

Playson will continue expanding to new customer bases and developing games, having architected a scalable, agile gaming service. Because availability is mission critical in the gaming industry, Playson is working toward deploying its infrastructure across multiple AWS Regions. "Using AWS, we can build a more stable service while providing the best experiences for players who are located in different regions," says Mylotskyi.

About Playson

Omnichannel casino game developer Playson has released 69 cutting-edge, high-quality gaming experiences, which the company delivers to thousands of iGaming businesses through its turnkey gaming service.

AWS Services Used

Amazon CloudFront

Amazon CloudFront is a content delivery network (CDN) service built for high performance, security, and developer convenience.

Learn more »

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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Amazon MemoryDB

Amazon MemoryDB for Redis is a durable database with microsecond reads, low single-digit millisecond writes, scalability, and enterprise security.

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Amazon DynamoDB

Amazon DynamoDB is a fully managed, serverless, key-value NoSQL database designed to run high-performance applications at any scale. DynamoDB offers built-in security, continuous backups, automated multi-Region replication, in-memory caching, and data import and export tools.

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